

Python For & While Loops - 30 Practice Questions with Answers

1. Print 1 to 10

```
for i in range(1,11): print(i)
```

2. Print 10 to 1

```
n=10
while n>=1:
    print(n)
    n-=1
```

3. First 20 even numbers

```
for i in range(2,41,2): print(i)
```

4. First 20 odd numbers

```
for i in range(1,40,2): print(i)
```

5. Sum 1 to N

```
n=int(input()); s=0
for i in range(1,n+1): s+=i
print(s)
```

6. Factorial

```
n=int(input()); f=1
for i in range(1,n+1): f*=i
print(f)
```

7. Multiplication table

```
n=int(input())
for i in range(1,11): print(n, 'x', i, '=', n*i)
```

8. Count digits

```
n=int(input()); c=0
while n>0: c+=1; n//=10
print(c)
```

9. Reverse number

```
n=int(input()); r=0
while n>0: r=r*10+n%10; n//=10
print(r)
```

10. Sum of digits

```
n=int(input()); s=0
while n>0: s+=n%10; n//=10
```

```
print(s)
```

11. Palindrome

```
n=int(input()); t=n;r=0
while n>0:r=r*10+n%10;n//=10
print('Palindrome' if r==t else 'Not')
```

12. Armstrong

```
n=int(input());t=n;s=0
while n>0:d=n%10;s+=d**3;n//=10
print('Armstrong' if s==t else 'Not')
```

13. Strong Number

```
n=int(input());t=n;s=0
while n>0:
    d=n%10;f=1
    for i in range(1,d+1):f*=i
    s+=f;n//=10
print('Strong' if s==t else 'Not')
```

14. Prime

```
n=int(input());c=0
for i in range(1,n+1):
    c+= n%i==0
print('Prime' if c==2 else 'Not')
```

15. Primes 1..N

```
n=int(input())
for x in range(2,n+1):
    c=0
    for i in range(1,x+1): c+=x%i==0
    if c==2: print(x)
```

16. Fibonacci

```
n=int(input());a,b=0,1
for _ in range(n): print(a);a,b=b,a+b
```

17. GCD

```
a=int(input());b=int(input())
while b:a,b=b,a%b
print(a)
```

18. LCM

```
a=int(input());b=int(input());x,y=a,b
while y:x,y=y,x%y
print(a*b//x)
```

19. Power

```
a=int(input());b=int(input());p=1
for _ in range(b):p*=a
print(p)
```

20. Largest digit

```
n=int(input());m=0
while n>0:m=max(m,n%10);n//=10
print(m)
```

21-25 Pattern programs

Practice star, reverse star, number and alphabet patterns using nested loops.

26. Count vowels

```
s=input();c=0
for ch in s.lower():
    c+= 1 if ch in 'aeiou' else 0
print(c)
```

27. Reverse string

```
s=input();r=''
for ch in s:r=ch+r
print(r)
```

28. Character frequency

```
s=input()
for ch in s:
    print(ch,s.count(ch))
```

29. Factors

```
n=int(input())
for i in range(1,n+1):
    if n%i==0: print(i)
```

30. Perfect number

```
n=int(input());s=0
for i in range(1,n):
    if n%i==0:s+=i
print('Perfect' if s==n else 'Not')
```